Sabancı University

Information Visualization Homework Report

Population Change and Socio-Economic Indicators in Türkiye (2019–2024)

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Project Question

What relationship exists between population growth in major Turkish cities and national socio-economic indicators such as unemployment rate, GDP per capita, and inflation between 2019 and 2024?

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1 Introduction

This report presents the design process and implementation for a visualization homework project based on Turkish population and socio-economic data. The study focuses on the years 2019 to 2024 and investigates whether changes in the populations of selected major cities move together with national indicators such as unemployment rate, GDP per capita, and inflation.

The report combines hand-drawn sketches, design rationale, and one coded visualization. Some sections intentionally contain placeholders so that additional handwritten images, screenshots, reflections, and course-specific terminology can be inserted later.

Main research question:

What relationship exists between population growth in major Turkish cities and national socio-economic indicators such as unemployment rate, GDP per capita, and inflation between 2019 and 2024?

2 Dataset Description

2.1 Scope of the Study

Although the original census dataset covered a broader time interval, only the years between 2019 and 2024 were used in this homework in order to align the population data with the socio-economic indicators collected for the same period.

The selected cities are:

- Van
- Muğla
- Adana
- Ankara
- İstanbul
- İzmir

2.2 Population Data Used

Table 1: Population of Selected Turkish Cities (2019–2024)

Year	Van	Muğla	Adana	Ankara	İstanbul	İzmir
2019	1,136,757	983,142	2,237,940	5,639,076	15,519,267	4,367,251
2020	1,149,342	1,000,773	2,258,718	5,663,322	15,462,452	4,394,694
2021	1,141,015	1,021,141	2,263,373	5,747,325	15,840,900	4,425,789
2022	1,128,749	1,048,185	2,274,106	5,782,285	15,907,951	4,462,056
2023	1,127,612	1,066,736	2,270,298	5,803,482	15,655,924	4,479,525
2024	1,118,087	1,081,867	2,280,484	5,864,049	15,701,602	4,493,242

2.3 Average Population Values Used in Some Sketches

For some sketches, an average city population value was used for simplicity and readability.

Table 2: Average Population Used in Simplified Sketches

Year	Average Population
2019	4,980,572
2020	5,021,550
2021	5,073,257
2022	5,100,555
2023	5,067,263
2024	5,089,889

2.4 Socio-Economic Indicators from TÜİK

The three national indicators selected from TÜİK for the same period are unemployment rate, GDP per capita (USD), and inflation rate.

Table 3: National Socio-Economic Indicators of Türkiye (2019–2024)

Year	Unemployment Rate	GDP per Capita (USD)	Inflation Rate
2019	13.7	9127	11.84
2020	13.2	8555	14.60
2021	12.0	9592	36.08
2022	10.4	10655	62.93
2023	9.4	13243	64.00
2024	8.6	15325	44.38

3 Task 1: Hand-Drawn Exploratory Sketches

3.1 Task 1: Türkiye Map with City-Based Time Series

In this task, a hand-drawn map of Türkiye was created, and the six selected cities were placed and labeled on the map. Next to each city, a small time-series line chart was drawn to show population change between 2019 and 2024.

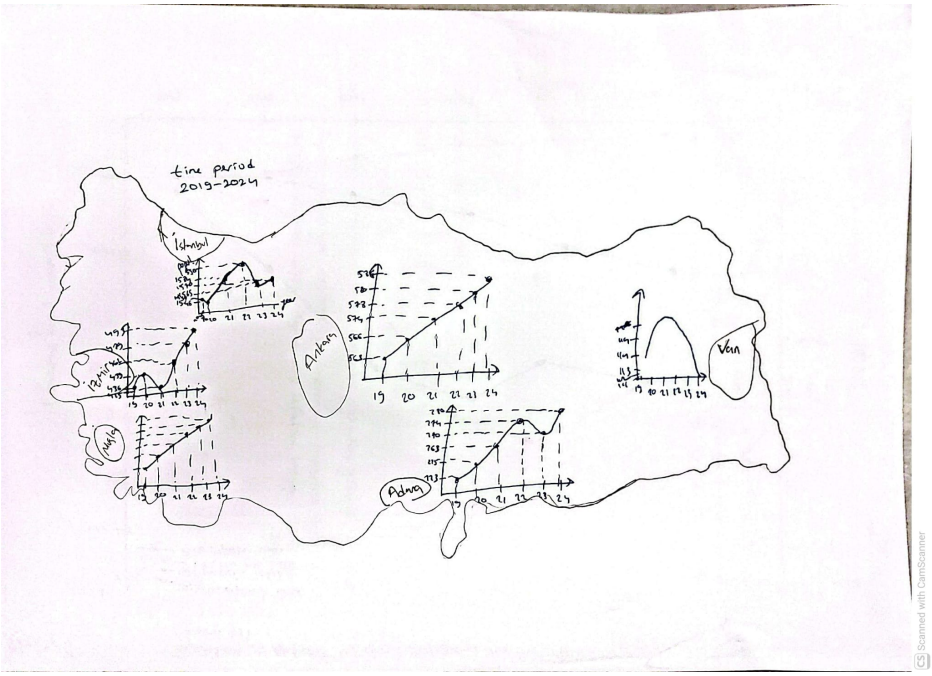


Figure 1: Hand-drawn Türkiye map with labeled cities and mini line charts (Task 1).

3.2 Task 1: Three Socio-Economic Indicators as Line Charts

In this task, the three national indicators were drawn as separate line charts on paper for the same time period (Unfortunately, the image did not fit on this page; please see the next page.).

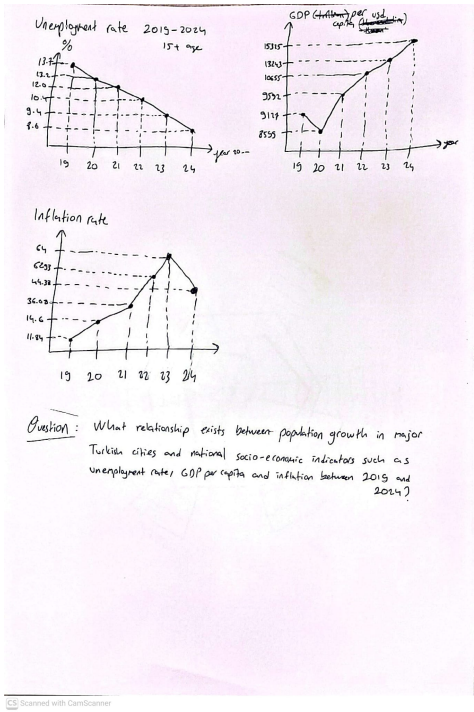


Figure 2: Hand-drawn line charts of unemployment rate, GDP per capita and inflation.

3.3 Task 1: Formulated Question

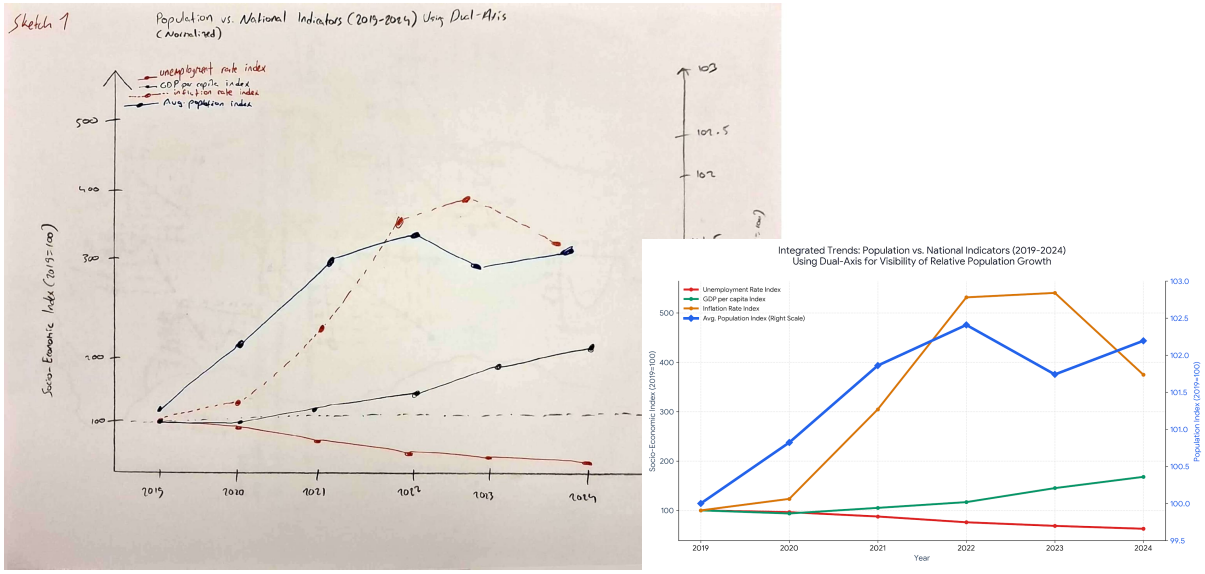
Based on Task 1, the following question was formulated:

> **What relationship exists between population growth in major Turkish cities and national socio-economic indicators such as unemployment rate, GDP per capita, and inflation between 2019 and 2024?**

Why this question?

This question allows us to visually explore possible relationships between urban population and socio-economic indicators such as unemployment, GDP per capita, and inflation

4 Task 2: Sketch 1: Multi-Line Comparison View



(a) Hand-drawn sketch of the visualization concept.

(b) Programmatically generated version used to clarify the design.

Figure 3: Multi-line comparison visualization showing the relationship between average population growth and national socio-economic indicators between 2019 and 2024.

Visualization type: Multi-line comparison chart with dual-axis scaling

Goal: The goal of this visualization is to compare temporal trends between population growth in major Turkish cities and national socio-economic indicators such as unemployment rate, GDP per capita, and inflation. By aligning all variables along the same timeline, the visualization helps reveal whether changes in economic conditions coincide with changes in population trends.

Design rationale:

This visualization uses a multi-line chart because it is effective for showing temporal patterns and comparing multiple variables over time. The horizontal axis encodes the time dimension (2019–2024), while vertical position encodes the magnitude of each variable. Different colors represent different indicators, allowing viewers to easily distinguish the trends. Lines and connected points act as visual marks that emphasize direction and slope, making increases or decreases clearly visible. A secondary axis is used for the population index so that variables with different scales can still be compared within the same view. This encoding enables quick identification of correlations, divergences, and turning points between demographic and economic indicators.

Where this could be useful:

This visualization could be useful for policymakers, economists, or researchers interested in understanding how demographic changes in major cities relate to broader economic conditions. It can also support exploratory analysis in educational or analytical settings.

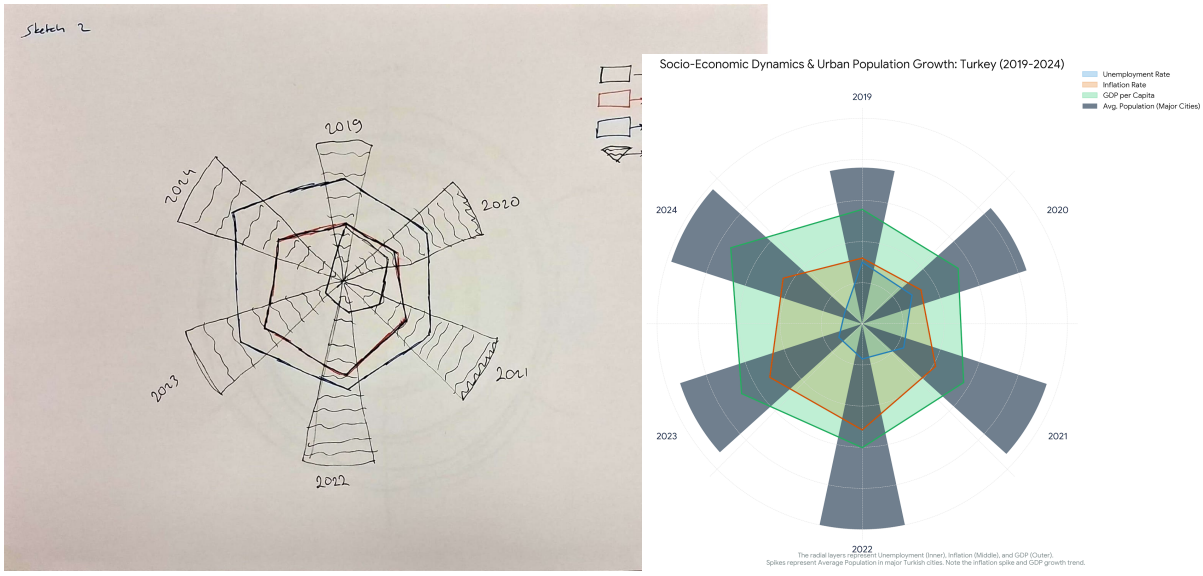
Strengths:

- Clearly communicates trends and changes over time.
- Enables direct comparison between multiple indicators.
- Makes it easier to visually detect possible correlations between variables.

Weaknesses:

- Dual-axis charts can sometimes be misinterpreted if scales are not carefully labeled.
- Too many variables could reduce readability and visual clarity.

4.1 Sketch 2: Radial Socio-Economic Pattern View



(a) Hand-drawn sketch of the radial visualization concept.

(b) Programmatically generated version used to clarify the design.

Figure 4: Radial visualization representing the relationship between socio-economic indicators and urban population trends across the years 2019–2024.

Visualization type: Radial (polar) multi-variable comparison chart

Goal: The goal of this visualization is to represent the evolution of socio-economic indicators and urban population trends in a circular timeline. By arranging the years around a radial axis, the visualization allows viewers to observe how different indicators change relative to each other across time.

Design rationale:

This visualization uses a radial layout to emphasize the cyclical and comparative nature of temporal data. Each year is encoded as an angular segment around the circle, while the magnitude

of each indicator is represented by radial distance from the center. Different indicators are encoded using color and line shapes, allowing viewers to distinguish between unemployment rate, inflation, GDP per capita, and average population. The radial bars represent the population index, while layered polygonal lines represent economic indicators. This design uses spatial position and color as visual channels to encode quantitative differences and enables comparison of patterns across years.

Where this could be useful:

This type of visualization can be useful in exploratory analysis or presentations where the goal is to highlight patterns or relationships between multiple indicators in a compact visual form. It may also help audiences quickly compare different years and identify unusual economic changes.

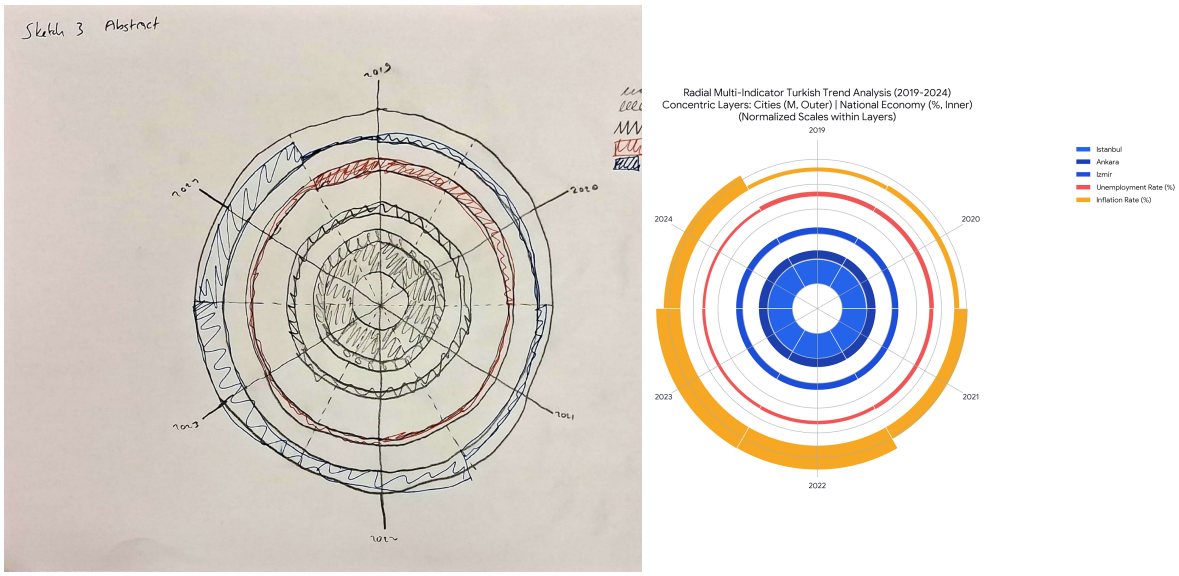
Strengths:

- Provides an intuitive overview of multiple variables in a single compact layout.
- Makes it easier to compare the relative magnitude of indicators across years.
- Visually distinctive representation that highlights patterns and spikes.

Weaknesses:

- Radial charts can be harder to interpret precisely compared to linear charts.
- Comparing exact values across different angles may be less intuitive for viewers.

4.2 Sketch 3: Abstract Concentric Trend Visualization



(a) Hand-drawn abstract sketch exploring layered temporal relationships.

(b) Programmatically generated abstract radial visualization.

Figure 5: Abstract concentric visualization combining population trends of major cities with national socio-economic indicators between 2019 and 2024.

Visualization type: Abstract concentric radial visualization with layered encoding

Goal: The goal of this visualization is to present the relationship between population growth in major Turkish cities and national economic indicators in a more conceptual and exploratory way. Instead of focusing on exact numeric comparison, the visualization emphasizes the overall patterns and relative differences between variables over time.

Design rationale:

This design uses a concentric radial structure where each circular layer represents a different data category. The inner layers encode national socio-economic indicators such as unemployment rate and inflation, while outer layers represent population-related measures for major cities. Time is encoded using angular segments around the circle, creating a circular timeline from 2019 to 2024. Variations in color, texture, and radial distance are used as visual channels to differentiate indicators and emphasize changes over time. This layered abstraction allows multiple datasets to be visually integrated without relying on traditional Cartesian axes, encouraging viewers to perceive relationships between economic conditions and demographic growth patterns.

Where this could be useful:

This type of visualization could be useful in exploratory analysis, storytelling dashboards, or presentations where the goal is to communicate overall patterns rather than precise values. It can also attract attention in interactive data visualizations where engaging visual forms help communicate complex multi-variable relationships.

Strengths:

- Integrates multiple datasets within a single compact visual structure.
- Encourages holistic pattern recognition rather than focusing only on precise values.
- Visually distinctive and engaging, which can help communicate complex relationships.

Weaknesses:

- Abstract representations may make precise quantitative comparison difficult.
- Interpretation may require additional explanation compared to traditional charts.

5 Task 3: Implemented Visualization

For the implementation stage, one of the three visualization concepts was selected and coded as a standalone HTML file with embedded JavaScript.

5.1 Chosen Visualization

Write which sketch you selected and why you selected it for implementation.

5.2 Implementation Details

- **Technology used:** *Example: HTML, CSS, JavaScript, D3.js*
- **File structure:** *Example: single standalone HTML file*
- **Interaction techniques used:** *Example: hover tooltips, highlighting, legends, labels*
- **Visual encoding:** *Brief summary of how variables are represented*

5.3 GitHub Repository

The implementation can be found at the following link:

https://github.com/your-username/your-repository

Replace the placeholder link above with your actual repository link.

5.4 Screenshots of the Final Visualization

Figure 6: Placeholder for screenshot of the implemented visualization.

Figure 7: Placeholder for an additional screenshot, such as interaction state, tooltip, or focused view.

5.5 Implementation Reflection

Write a short reflection about whether the coded version successfully answered the research question, what worked well, and what could still be improved.

6 Discussion

In this section, summarize the main visual patterns you observed after sketching and implementation. For example, mention whether some cities show steady growth while others fluctuate, and whether those changes appear to move together with unemployment, GDP per capita, or inflation. Keep this section interpretive rather than purely descriptive.

7 Conclusion

This homework explored the relationship between urban population change and national socio-economic indicators in Türkiye between 2019 and 2024. The project involved three stages: exploratory sketching, concept generation, and coded implementation. Through these stages, the report aimed to move from raw tabular values toward visual forms that support comparison, trend analysis, and interpretation.

Add 1 short paragraph here summarizing your final takeaway.

Appendix A: Optional Source Image Placeholder

If you want to include the source screenshot that contains the socio-economic indicator values, you may place it here.

Figure 8: Placeholder for the source image containing unemployment, GDP per capita, and inflation values.

Optional note: If the image file is stored locally, replace the placeholder above with a command such as:

```
\includegraphics[width=0.85\textwidth]{tuik_indicators.png}
```

Appendix B: Practical Notes for Replacing Placeholders

- To replace an image placeholder, use:

  ```
  \includegraphics[width=0.9\textwidth]{your-image-file-name}
  ```

- For large hand-drawn sketches, using `[width=0.8\textwidth]` or `[width=0.9\textwidth]` usually works well.
- If you want to place two screenshots side by side, use the `subfigure` environment from the `subcaption` package.
- If your GitHub repository contains a single HTML file, mention that it can be opened directly in a browser.